

Bacillary Angiomatosis — Comprehensive Disease Characterization Report

Disease: Bacillary Angiomatosis (BA) **MONDO:** MONDO:0000758 · **Category:** Infectious disease (bacterial, vasoproliferative) **Report basis:** Aggregated disease-level evidence from peer-reviewed literature (reviews, landmark case-control study, mechanistic in vitro/in vivo studies). No individual-patient/EHR dataset was provided; findings are synthesized from PubMed primary literature. Evidence types are flagged as *human clinical*, *in vitro*, or *model organism*.

Summary (Answer to the Research Question)

Bacillary angiomatosis is an angioproliferative (vascular tumor-forming) infectious disease caused by the Gram-negative bacteria *Bartonella henselae* and *Bartonella quintana*, occurring almost exclusively in immunocompromised hosts—most classically advanced HIV/AIDS (CD4 typically <100 cells/μL) and also solid-organ transplant recipients—and only rarely in immunocompetent individuals. The bacteria hijack vascular endothelium: the secreted autotransporter **BafA** acts as a VEGF-receptor-2 agonist, the trimeric autotransporter adhesin **BadA** activates HIF-1 to drive proangiogenic reprogramming, and the **VirB/D4 type IV secretion system** injects **Bep effectors** that inhibit endothelial apoptosis—together producing lobular capillary proliferations in skin and viscera. It is diagnosed by biopsy (Warthin-Starry silver stain, PCR) and is highly curable with prolonged antibiotics (erythromycin or doxycycline), but disseminated untreated disease can be fatal. It is **not a genetic disease**; there are no causal human genes, and host immunosuppression plus species-specific exposures (cats/cat fleas for *B. henselae*; homelessness, alcoholism, body lice for *B. quintana*) are the key determinants.

1. Disease Information

Overview. BA is a rare, treatable, vasoproliferative bacterial infection characterized by tumor-like proliferations of small blood vessels in skin, subcutaneous tissue, and internal organs. First recognized in the HIV/AIDS epidemic of the 1980s, it is caused by chronic infection with *Bartonella henselae* or *B. quintana* (formerly genus *Rochalimaea*)

([P 21285862](#)

[P 8335458](#)

Key identifiers. - **MONDO:** MONDO:0000758 - **MeSH:** Angiomatosis, Bacillary (D016917) - **ICD-10:** A44.8 (Other forms of bartonellosis); **ICD-11:** 1C30.Y (Other specified bartonellosis) - **OMIM / Orphanet:** Not a Mendelian disease → no OMIM gene entry; not a classic Orphanet rare-genetic entry (infectious etiology) - **SNOMED CT:** Bacillary angiomatosis (disorder)

Synonyms / alternative names: Bacillary angiomatosis-peliosis (when visceral peliosis coexists); epithelioid angiomatosis; bacillary epithelioid angiomatosis; (historically) *Rochalimaea* angiomatosis. Bacillary peliosis (hepatis/splenic) is the visceral counterpart.

Information source. Disease-level aggregated resources and primary literature (case series, one landmark molecular case-control study, mechanistic studies), not individual EHR data.

2. Etiology

Causal factors — infectious (not genetic). BA is caused by two Gram-negative bacilli: *Bartonella henselae* (NCBI:txid38323) and *Bartonella quintana* (NCBI:txid803). "*Bacillary angiomatosis is an infectious disease caused by 2 gram-negative bacilli, Bartonella henselae and Bartonella quintana*"

([P 21285862](#)

A permissive host state (immunosuppression) determines progression from infection to vasoproliferation: "*The severity of Bartonella infection correlates with the patient's immune status*"

([P 24933445](#)

Risk factors (environmental / host). - **Immunosuppression (dominant):** advanced HIV/AIDS (e.g., CD4 47 cells/ μ L,

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 solid-organ transplant recipients
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 Rare in immunocompetent hosts
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- ***B. henselae* exposures:** cat contact and cat-flea (*Ctenocephalides felis*) exposure ($P \leq 0.004$)

),
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- ***B. quintana* exposures:** low income ($P=0.003$), homelessness ($P=0.004$), body-louse exposure ($P=0.03$)

);
 chronic alcoholism
 (

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- **Genetic risk factors:** None known. **Not applicable** — no susceptibility loci, causal variants, or modifier genes are established for the human host.

Protective factors. Immune competence / immune reconstitution (antiretroviral therapy in HIV) reduces risk and aids cure; ectoparasite control (cat flea control, delousing/hygiene) reduces exposure. No genetic protective variants are described.

Gene–environment interactions. Not applicable in the classical (human-genetic) sense; the operative interaction is **host immune status** $\times$ **pathogen exposure** (e.g., cat/flea or louse contact in an immunosuppressed host).

3. Phenotypes (with suggested HPO terms and qualitative frequencies)

BA phenotypes are *clinical signs/physical manifestations* (cutaneous vascular lesions), *symptoms* (fever, malaise), and *laboratory abnormalities* (anemia, thrombocytopenia). Onset is adult (tracks the immunosuppressed adult population); severity ranges mild-to-severe; course is progressive if untreated, resolving with therapy.

Phenotype	Type	Suggested HPO	Frequency/notes
Cutaneous/subcutaneous vascular papules & nodules (red-violaceous, friable, bleed when traumatized)	Physical sign	HP:0200039 (Papule); HP:0000988 (Skin nodule); HP:0011276 (Vascular skin abnormality)	Most common; hallmark manifestation (P 21285862 , P 11362939)
Fever	Symptom	HP:0001945	Frequent, esp. systemic disease (P 10718405)
Lymphadenopathy	Sign	HP:0002716	Common (P 22316326)
Hepatomegaly / peliosis hepatis	Sign/organ	HP:0002240; HP:0410042 (Hepatic peliosis)	<i>B. henselae</i> (P 9407154)
Splenomegaly / splenic peliosis	Sign	HP:0001744	Visceral disease (P 22316326)
Lytic bone lesions / bone pain	Sign	HP:0002754 (Osteomyelitis)/ HP:0002653; HP:0002917	<i>B. quintana</i> , long bones (P 9407154 , P 11362939)
Weight loss / anorexia / malaise	Symptom	HP:0001824; HP:0002039	Systemic disease (P 10718405)
Anemia	Lab	HP:0001903	Severe cases (P 18098054)
Thrombocytopenia	Lab	HP:0001873	

Phenotype	Type	Suggested HPO	Frequency/notes
			<i>B. quintana</i> bacteremia (P 9895398)
GI involvement (hematemesis, mucosal nodules)	Sign	HP:0002248; HP:0025314	Rare (P 12894361)
Oral cavity lesions	Sign	HP:0000155 (Oral mucosa abnormality)	Rare (P 28902296), (P 10718405)
Panserositis (pleuritis/pericarditis/peritonitis)	Sign	HP:0032261/HP:0001698	Rare severe (P 18098054)

Quality-of-life impact: Painful/disfiguring skin lesions, bleeding, and systemic constitutional symptoms impair function; visceral disease can be life-threatening. No formal EQ-5D/SF-36 data exist for BA specifically; QoL burden is inferred from lesion morbidity and treatable nature.

4. Genetic / Molecular Information

Not applicable to the human host — BA is a bacterial infection, not an inherited disorder. There are **no causal human genes, pathogenic germline/somatic variants, ClinVar entries, allele frequencies, modifier genes, epigenetic disease signatures, or chromosomal abnormalities.**

The relevant "molecular" players are **bacterial virulence genes** (host-microbe interaction): - *bafA* — proangiogenic autotransporter; passenger domain functions as a VEGF analog
([P 32678094](#));
homologs in *B. quintana*, *B. bacilliformis*, *B. elizabethae*
([P 35379004](#)),

- 36810719).
- **badA** (Bartonella adhesin A) — trimeric autotransporter adhesin; head domain mediates host adhesion and HIF-1 activation (18627378).
 - **virB/virD4** T4SS and **Bep** effector genes — translocation of effectors inhibiting apoptosis and driving proangiogenic phenotype (21557057).
 - *B. quintana* Vmps (variably expressed outer-membrane proteins) and Trw conjugation system for erythrocyte adherence (23163798).
 - *B. quintana* Vmps (variably expressed outer-membrane proteins) and Trw conjugation system for erythrocyte adherence (21557057).

5. Environmental Information

- **Infectious agents (causal):** *Bartonella henselae* and *Bartonella quintana* (Alphaproteobacteria; family Bartonellaceae). *B. henselae* also causes cat scratch disease and peliosis hepatis; *B. quintana* also causes trench fever and endocarditis (18444576).

24933445).

A rare BA case from *B. elizabethae* has been reported

36810719).

- **Vectors/reservoirs:** *B. henselae* — reservoir cats, vector cat flea *Ctenocephalides felis*; *B. quintana* — reservoir humans, vector body louse *Pediculus humanus corporis* (9272384).

- **Lifestyle factors:** homelessness, poverty, chronic alcoholism (for *B. quintana*); cat ownership/exposure (for *B. henselae*)

(
P 9407154
P 7529895
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- **Toxins/radiation/occupational exposures:** Not applicable.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating infection → clinical lesion)

1. **Exposure/inoculation** — a cat scratch/flea (*B. henselae*) or body louse (*B. quintana*) **introduces** *Bartonella* into skin/blood of a (usually immunocompromised) host. (*human clinical/epidemiologic*;

P 9407154

2. **Bacterial adhesion** — the trimeric autotransporter adhesin **BadA binds** fibronectin/ECM and endothelial cells, **leading to** autoagglutination and firm attachment. (*in vitro*;

P 18627378

P 21557057

3. **Proangiogenic reprogramming (branch A)** — BadA engagement **activates HIF-1**, which **results in** transcription of angiogenic genes and **autocrine VEGF secretion** by endothelial cells. (*in vitro*;

P 18627378

P 23163798

4. **VEGFR-2 mitogenic signaling (branch B)** — the secreted autotransporter **BafA binds** **VEGF receptor-2** and **activates** its downstream pathway (functioning as a VEGF analog), **driving** endothelial proliferation, tube formation, and microvessel sprouting. (*in vitro* + *mouse*;

P 32678094

5. **Anti-apoptotic survival (branch C)** — the **VirB/D4 T4SS translocates Bep effectors** into endothelial cytoplasm, **causing** invasome-mediated uptake of bacterial aggregates and

inhibiting apoptosis, so proliferating endothelium survives. (*in vitro*;

(P 23163798)

6. **Stromal amplification** — infected **pericytes** increase VEGF with reduced coverage

(P 23184416),

and persistence in **mesenchymal stromal cells** further **amplifies** endothelial activation/angiogenesis

(P 34031126).

(*in vitro*)

7. **Lesion formation** — combined proliferation + survival + inflammation **produces** a lobular capillary proliferation lined by plump epithelioid endothelial cells with neutrophilic infiltrate and stromal bacterial aggregates — the histologic **bacillary angioma**. (*human clinical*;

(P 24718378)

8. **Dissemination** — because host immunity is impaired, lesions **spread** hematogenously to viscera (liver/spleen peliosis, bone, GI tract, CNS), which **can lead to** organ dysfunction, hemorrhage, and death if untreated. (*human clinical*;

(P 11362939),

(P 24933445)

Upstream vs downstream: Adhesion (BadA) and effector delivery (VirB/D4) are upstream; VEGF/VEGFR-2 signaling, HIF-1 activation, and apoptosis inhibition are the core intermediate nodes; endothelial proliferation → lesion formation → dissemination are downstream.

Molecular pathways: VEGF/VEGFR-2 signaling, HIF-1 α transcriptional program, NF- κ B-associated proangiogenic/anti-apoptotic signaling. **KEGG/Reactome:** VEGF signaling pathway; HIF-1 signaling pathway. **Cellular processes:** angiogenesis, endothelial proliferation, inhibition of apoptosis, inflammation. **Immune involvement:** disease is permitted by **immunodeficiency**; lesions show mixed neutrophilic/lymphocytic infiltrate; in immunocompetent hosts a granulomatous response contains infection

(P 9784568).

Suggested GO terms: angiogenesis (GO:0001525); sprouting angiogenesis (GO:0002040); VEGF receptor signaling pathway (GO:0048010); cellular response to hypoxia (GO:0071456); negative

regulation of apoptotic process (GO:0043066); inflammatory response (GO:0006954). **Suggested CL terms:** endothelial cell (CL:0000115); blood vessel endothelial cell (CL:0000071); pericyte (CL:0000669); mesenchymal stem cell (CL:0000134); erythrocyte (CL:0000232, for *B. quintana* bacteremia). **Omics:** No human disease transcriptomic/proteomic/metabolomic BA signatures are established; mechanistic data are protein/pathway-level (VEGF, HIF-1, BafA/BadA/Bep).

7. Anatomical Structures Affected

- **Primary organ:** skin (UBERON:0002097) and subcutaneous tissue (UBERON:0002072) — cutaneous/subcutaneous vascular nodules.
 - **Secondary/visceral organs:** liver (UBERON:0002107; peliosis hepatis), spleen (UBERON:0002106), bone/bone marrow (UBERON:0002481/UBERON:0002371; lytic lesions, long bones), lymph nodes (UBERON:0000029), GI tract (UBERON:0000160; esophagus/stomach/duodenum), oral mucosa (UBERON:0003343), lung (UBERON:0002048), heart/pericardium, CNS/brain (UBERON:0000955)
- (
- P 11362939
- P 12894361
- P 16495874
-).
- **Body systems:** primarily **cardiovascular/vascular** (endothelium), with **integumentary, hepatobiliary, skeletal, lymphatic/hematologic, digestive**, and occasionally **nervous** involvement.
 - **Tissue/cell level:** vascular (connective/endothelial) tissue; target cells = **vascular endothelial cells** (CL:0000115), with **pericytes** (CL:0000669) and stromal cells; *B. quintana*/*B. henselae* also infect **erythrocytes** (CL:0000232) causing bacteremia.
 - **Subcellular:** bacteria localize extracellularly (stromal aggregates) and are taken up intracellularly via the **invasome**; effectors act in the endothelial **cytoplasm/plasma membrane** (GO:0005886 plasma membrane; GO:0005829 cytosol).
 - **Lateralization:** typically multifocal/bilateral and disseminated rather than lateralized; cutaneous lesions are widely distributed.

8. Temporal Development

- **Onset:** adult-onset (parallels immunosuppressed adults); **subacute/chronic/insidious** course over weeks to months (e.g., 6-month growing lesions,

(P 23282705

10-month nodule,

) P 9205511

- **Progression:** **progressive** if untreated, from localized papules to disseminated visceral disease; can be **rapidly life-threatening** with hemorrhage/organ involvement

(P 12894361

) P 18098054

- **Course pattern:** treatment-induced remission is the rule with adequate therapy; **relapse** occurs with inadequate/short courses

(P 9272384

) P 9205511

- **Duration:** not self-limited in immunocompromised hosts (requires antibiotics); may resolve with prolonged therapy and immune reconstitution.

- **Critical intervention window:** early biopsy-based diagnosis and prompt antibiotics prevent dissemination and death; immune restoration (ART) is a key adjunct.

9. Inheritance and Population

- **Epidemiology:** Rare; no precise population prevalence/incidence figures exist (opportunistic, under-reported). Incidence fell sharply in high-income settings after combination antiretroviral therapy reduced severe HIV immunosuppression; it remains under-recognized in sub-Saharan Africa despite high HIV prevalence

(P 24718378

) *B. quintana* seroprevalence is high in homeless populations (IgG $\geq 1:128$ in 57% and $\geq 1:1024$

in 11% of Tokyo homeless;

(P 16495631);

B. quintana bacteremia in 14% of Marseille homeless

(P 9895398).

- **Inheritance: Not applicable** (infectious disease; no Mendelian inheritance, penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity, or carrier frequency).
- **Demographics:** predominantly adults with immunosuppression; historically HIV-infected men; *B. quintana* clusters in homeless/alcoholic, low-income urban populations. No strong ethnic predisposition beyond socioeconomic exposure patterns. Sex ratio reflects the underlying at-risk populations rather than intrinsic disease bias.
- **Geographic distribution:** worldwide; *B. quintana* ("urban trench fever") in homeless populations across the US, Europe, and Japan

(P 11871479),

(P 16495631).

10. Diagnostics

- **Biopsy/histopathology (gold standard):** lobular proliferation of capillaries lined by plump, epithelioid/spindled endothelial cells; mixed neutrophilic/leukocytoclastic infiltrate; extracellular amphophilic granular material (bacterial aggregates)

(P 24718378),

(P 12152480).

Warthin-Starry silver stain highlights clumps of bacilli — "A Warthin-Starry stain highlighted clumps of bacilli, confirming the diagnosis of BA"

(P 24718378).

Electron microscopy can visualize bacilli

(P 11362939).

- **Molecular:** PCR of lesional tissue (e.g., 16S rRNA, citrate synthase *glbA*, riboflavin synthase *ribC*, or a 298-bp *Bartonella* fragment) identifies and speciates the organism and can **monitor treatment response**

(
P 9205511,
P 7529895,
P 40090362).)

- **Serology:** indirect immunofluorescence antibody (IFA) for *Bartonella*-specific IgG/IgM; useful epidemiologically but **poorly reliable** for individual BA diagnosis

(
P 9272384,
P 11362939).)

- **Culture:** possible but **difficult/fastidious**, slow-growing; low sensitivity

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P 9272384).)

- **Imaging:** CT/MRI/ultrasound to define visceral (hepatosplenic peliosis, abscesses) and osseous lesions

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P 27428207,
P 22316326).)

- **Laboratory:** anemia, thrombocytopenia, elevated inflammatory markers may accompany systemic disease.

- **Genetic/omics testing:** **Not applicable** (no human genetic test; no karyotype/CMA/mtDNA/ repeat-expansion role).

- **Differential diagnosis:** Kaposi sarcoma (key mimic), pyogenic granuloma, angiosarcoma, epithelioid hemangioma, verruga peruana (*B. bacilliformis*), cat-scratch disease — distinguished by Warthin-Starry-positive bacilli and vascular-proliferative histology

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P 10718405,
P 31780437).)

BA is antibiotic-curable, unlike KS.

- **Screening:** No population screening; consider *B. quintana* testing in symptomatic homeless/ alcoholic patients and BA in HIV patients with vascular skin lesions.

11. Outcome / Prognosis

- Survival/mortality:** With appropriate prolonged antibiotics, prognosis is **excellent** (lesions resolve). **Untreated Bartonella infection can cause high mortality** ([P 24933445](#)); disseminated/visceral disease (e.g., GI hemorrhage, panserositis) is potentially fatal ([P 12894361](#) , [P 18098054](#)).
- Morbidity:** disfiguring/painful, friable, bleeding skin lesions; visceral organ dysfunction; bone pain/lytic lesions. No standardized disability/QoL metrics for BA.
- Disease course/complications:** hemorrhage from friable lesions, airway/oronasal involvement (ornasal fistula, [P 28902296](#)), massive GI bleeding ([P 12894361](#)), anemia and multi-serosal involvement ([P 18098054](#)).
- Recovery potential:** high with treatment + immune reconstitution; relapse if therapy is too short.
- Prognostic factors:** degree/duration of immunosuppression (CD4 count), extent of dissemination, timeliness of diagnosis and adequacy/duration of antibiotics. PCR negativity of lesional material is a favorable treatment-response marker ([P 9205511](#)).

12. Treatment

- **First-line pharmacotherapy: Erythromycin** (macrolide) is first-line for angioproliferative lesions — *"erythromycin is the first-line antibiotic therapy for the treatment of angioproliferative lesions"*
([P 24933445](#)).
- **Doxycycline** (tetracycline) is the principal alternative — *"Doxycycline led to complete resolution"*
([P 28902296](#)).
- Azithromycin and clarithromycin are also effective
([P 21285862](#)),
([P 18098054](#)).
- **Duration:** prolonged — **erythromycin for at least three months** is recommended (or doxycycline) to prevent relapse in immunocompromised patients
([P 11362939](#)).
- **Severe/systemic disease:** add **rifampicin; gentamicin + doxycycline** for associated bacteremia/endocarditis
([P 24933445](#)).
- **Adjuncts: antiretroviral therapy** for immune reconstitution in HIV patients
([P 27428207](#));
treat/eliminate ectoparasites.
- **Surgical/interventional:** localized excision has been used but relapse/dissemination risk means antibiotics are essential
([P 12152480](#)),
([P 16495874](#)).
- **Advanced/experimental therapeutics (gene/cell/RNA/targeted/immunotherapy): Not applicable** — antibiotics are curative; no such therapies are used.

- **Adverse events:** relate to the antibiotics used (GI intolerance with erythromycin/doxycycline; photosensitivity with doxycycline); a Jarisch-Herxheimer-like reaction can occur at treatment initiation.
- **Suggested NCIT/CHEBI terms:** Erythromycin (NCIT:C578; CHEBI:42355); Doxycycline (NCIT:C513; CHEBI:50845); Azithromycin (NCIT:C609; CHEBI:2955); Clarithromycin (CHEBI:3732); Rifampicin (NCIT:C29408; CHEBI:28077); Gentamicin (NCIT:C61796; CHEBI:27412).

13. Prevention

- **Primary prevention:** avoid the exposures — for *B. henselae*: cat-flea control, prompt cleaning of cat scratches, caution around cats (esp. for immunosuppressed)
([P 22316326](#));
for *B. quintana*: **body-louse control/delousing, hygiene, and addressing homelessness**
([P 9895398](#)),
([P 11871479](#)).
- **Secondary prevention:** early recognition and biopsy of vascular skin lesions in immunosuppressed/homeless patients; prompt antibiotics prevent dissemination.
- **Tertiary prevention:** adequate-duration antibiotics + immune reconstitution (ART) to prevent relapse and complications.
- **Immunization:** None available (no vaccine).
- **Public-health interventions:** vector control (fleas/lice), sanitation and services for homeless populations, clinician awareness.
- **Genetic counseling/screening:** Not applicable (non-genetic).
- **Prophylaxis:** no established antibiotic prophylaxis; risk-factor reduction is the mainstay.

14. Other Species / Natural Disease

- **Taxonomy of pathogens:** *Bartonella henselae* (NCBI:txid38323), *Bartonella quintana* (NCBI:txid803); related agents of pathological angiogenesis include *B. bacilliformis* (verruca peruana; NCBI:txid774) and *B. elizabethae* (NCBI:txid34113)

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- **Natural reservoir/animal disease:** Cats (*Felis catus*, NCBI:txid9685) are the natural reservoir of *B. henselae*, harboring asymptomatic intraerythrocytic bacteremia — they are typically **infected but not diseased** (do not develop BA)

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P 18444576
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P 21637717

Fleas (*Ctenocephalides felis*) and, for *B. quintana*, human body lice are vectors/reservoirs

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P 35545848
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P 40090362

- **Comparative biology:** *B. henselae* induces angiogenesis in **human** endothelial cells but **not feline** endothelial cells, highlighting host-specific susceptibility

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P 21637717
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Bartonella is the only bacterial genus known to induce pathological angiogenesis in mammals

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P 23184416
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- **Zoonotic potential:** *B. henselae* is a **zoonosis** (cat-to-human via scratch/flea); *B. quintana* is human-adapted (louse-borne, human reservoir)

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P 9272384
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No orthologous human "disease gene" applies (bacterial etiology).

- **VBO breed associations:** Not applicable.

15. Model Organisms

- **In vitro (primary models):** human umbilical vein and skin microvascular **endothelial cell** lines (accelerated angiogenesis/wound healing on *B. henselae* infection;

(P 21637717);

human brain vascular **pericytes** (increased VEGF;

(P 23184416);

mesenchymal stromal cells (persistence amplifying angiogenesis;

(P 34031126);

recombinant **BafA passenger domain** assays (EC proliferation, tube formation, sprouting;

(P 32678094).

- **Mouse model:** immunocompetent **C57BL/6** mice given intraperitoneal *B. henselae* develop **granulomatous hepatitis** peaking week 4, resolving by 12 weeks, with hepatic bacterial DNA persistence ≥ 3 months

(P 9784568).
(model organism)

- **In vivo angiogenesis:** BafA drives **angiogenesis in mice**

(P 32678094).

- **Genetic/transgenic disease models: Not applicable** — BA is an infection; there are no knockout/knock-in "BA models." Bacterial mutants (e.g., *badA*⁻, *bafA* transposon mutants, *virB/D4* strains) are used to dissect virulence

(P 18627378),

(P 32678094),

(P 23163798).

- **Phenotype recapitulation & limitations:** cellular models reproduce the **angiogenic/anti-apoptotic signaling**; the mouse model reproduces **granulomatous containment** in an immunocompetent host but **not** the disseminated vasoproliferative tumors of immunocompromised humans — underscoring that human immunosuppression is essential to full disease.

Supported and Refuted Hypotheses

Supported: 1. BA is caused by *B. henselae* and *B. quintana* and occurs mainly in immunosuppressed hosts.

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2. Vasoproliferation is driven by bacterial proangiogenic factors: BafA (VEGFR-2 agonist), BadA (HIF-1 activation), and VirB/D4-delivered Beps (apoptosis inhibition).

(P 32678094

) P 18627378

) P 23163798

3. The two species have partly distinct tropism/exposures: *B. quintana* → subcutaneous/lytic bone lesions, homeless/louse-associated; *B. henselae* → peliosis hepatis, cat/flea-associated.

(P 9407154

4. Diagnosis rests on Warthin-Starry histopathology + PCR; treatment with prolonged macrolide/tetracycline is curative.

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Refuted / excluded: - BA is **not** a genetic/heritable disease; no human causal genes, variants, or inheritance pattern apply. - BA is **not** a neoplasm despite tumor-like appearance; it is a reversible, antibiotic-curable infection (distinguishing it from its mimic Kaposi sarcoma).

Limitations and Future Directions

- Epidemiology lacks precise prevalence/incidence rates (rare, opportunistic, under-reported); QoL and formal outcome metrics are absent for BA specifically.

- Mechanistic data derive largely from in vitro human EC/pericyte systems and *B. henselae*; fewer *B. quintana* mechanistic studies exist, and no immunocompromised animal model fully reproduces disseminated human BA.
 - Future work: an immunodeficient animal model of disseminated BA; comparative *B. quintana* vs *B. henselae* proangiogenic effector biology; molecular epidemiology in understudied high-HIV-burden regions.
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Evidence base: ~15 primary references (reviews, one landmark NEJM molecular case-control study [PMID 9407154], mechanistic in vitro/in vivo studies). Evidence types span human clinical (case series/epidemiology), in vitro (endothelial/pericyte/effector studies), and model organism (mouse). No individual-patient dataset was provided.

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